Homework 1, 26.10.2026

HOMEWORK 2.2

Exercise 2.2 Show that $\mathcal{C} = \sigma(\mathcal{C})$ if and only if $\mathcal{C}$ is a sigma algebra.

$\mathcal{B}(C) = C \iff C$ is a $\mathcal{B}$-algebra

($\Rightarrow$) definition of the generated algebra from a set

($\Leftarrow$) C is a $\mathcal{B}$-algebra, $\mathcal{B}(C)$ is the smallest $\mathcal{B}$-algebra containing all elements from C

$\mathcal{F} = \cap_i \mathcal{F}_i$, $\mathcal{F}_i$ is $\mathcal{B}$-algebra, $C \in \mathcal{F}_i \Rightarrow \mathcal{F} = \mathcal{B}(C)$

$\Rightarrow \mathcal{F}_i = C \cup D \Rightarrow C \subseteq \mathcal{B}(C)$

Now let's look at $\mathcal{F}_j = C$ (it exists because C is a $\mathcal{B}$-algebra)

$\Rightarrow \mathcal{F}_j$ is the smallest $\Rightarrow \mathcal{B}(C) \overset{\text{def.}}{=} \mathcal{F} = C$

HOMEWORK 2.3

Exercise 2.3 Let $\mathcal{C}$ and $\mathcal{D}$ be two collections of subsets on Ω such that $\mathcal{C} \subset \mathcal{D}$. Prove that $\sigma(\mathcal{C}) \subseteq \sigma(\mathcal{D})$.

$C, D \subseteq \Omega$, $C \subset D$ Prove $\mathcal{B}(C) \subseteq \mathcal{B}(D)$

What we know: $C \in \mathcal{B}(C)$, $D \in \mathcal{B}(D) \overset{C \subset D}{\Rightarrow} C \in \mathcal{B}(D)$

let: $\mathcal{B}(C) = \cap_i \mathcal{F}_i$, where $\{\mathcal{F}_i\}_i$ collection of all $\mathcal{B}$-algebras containing C

$\rightsquigarrow$ since $\mathcal{B}(D)$ is such $\Rightarrow \exists j : \mathcal{F}_j = \mathcal{B}(D) \Rightarrow \mathcal{B}(C) = \left(\bigcap_{i \neq j} \mathcal{F}_i\right) \cap \mathcal{F}_j \subseteq \mathcal{B}(D)$